

Classification of Cancer outcome using Genetic and Clinical data

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Exploratory Data Analysis

Introduction

This project investigates the effect of genetic and clinical variables on the survival outcome of breast cancer patients. The data contains records of more than 1000 breast cancer patients from several research institutions. Clinical data contains patient-related and tumor-related information. Additionally mRNA gene expression data is available for each patient. The gene expression data has been processed to include only the top 5000 most variable genes on the transformed scale ($\log_2(\text{counts} + 1)$).

We consider that the main outcome variable of interest is **vital_status**, which is defined in clinical data.

Load data

```
load("mrr_bio.Rdata")
```

Genes data

```
dim(GeneX)
```

```
## [1] 1231 5000
```

```
str(GeneX)
```

```
## num [1:1231, 1:5000] 0 0 10.92 3.17 5 ...
```

```
## - attr(*, "dimnames")=List of 2
```

```
## ..$ : chr [1:1231] "EW-A2FS-01A-11R-A17B" "OL-A6VR-01A-32R-A33J" "E9-A226-01A-21R-A157" "A8-A08H-01A-21R-A00Z"
```

```
## ..$ : chr [1:5000] "CLEC3A" "SCGB2A2" "CPB1" "GSTM1" ...
```

```
colnames(GeneX)[1:5]
```

```
## [1] "CLEC3A" "SCGB2A2" "CPB1" "GSTM1" "TFF1"
```

```
rownames(GeneX)[1:5]
```

```
## [1] "EW-A2FS-01A-11R-A17B" "OL-A6VR-01A-32R-A33J" "E9-A226-01A-21R-A157"
```

```
## [4] "A8-A08H-01A-21R-A00Z" "D8-A27H-01A-11R-A16F"
```

GeneX is a table with 1231 rows and 5000 columns:

- The rows correspond to the patients with each row name being the patient identifier such as EW-A2FS-01A-11R-A17B.

- The columns correspond to the genes with each column name being a gene such as `CLEC3A`, which are the 5000 most variable genes from one person to another.
- Each cell corresponds to the expression value (a float) of a certain gene (column) for a certain patient (row) -> continuous variables

Clinical data

```
load("mrr_bio.Rdata")
dim(clinical_data)
```

```
## Loading required namespace: S4Vectors
```

```
## NULL
```

```
head(clinical_data[, 1:5])
```

```
## DataFrame with 6 rows and 5 columns
##           initial_weight tissue_type laterality
##           <numeric> <character> <character>
## EW-A2FS-01A-11R-A17B           110      Tumor      Left
## OL-A6VR-01A-32R-A33J           380      Tumor      Right
## E9-A226-01A-21R-A157           510      Tumor      Left
## A8-A08H-01A-21R-A00Z           250      Tumor      Left
## D8-A27H-01A-11R-A16F           170      Tumor      Right
## D8-A3Z6-01A-11R-A239            60      Tumor      Left
##           tissue_or_organ_of_origin age_at_diagnosis
##           <character>           <integer>
## EW-A2FS-01A-11R-A17B      Breast, NOS           15302
## OL-A6VR-01A-32R-A33J      Breast, NOS           17702
## E9-A226-01A-21R-A157      Breast, NOS           16511
## A8-A08H-01A-21R-A00Z      Breast, NOS           24137
## D8-A27H-01A-11R-A16F      Breast, NOS           26588
## D8-A3Z6-01A-11R-A239      Breast, NOS           20629
```

```
colnames(clinical_data)
```

```
## [1] "initial_weight"           "tissue_type"
## [3] "laterality"               "tissue_or_organ_of_origin"
## [5] "age_at_diagnosis"         "primary_diagnosis"
## [7] "prior_treatment"          "diagnosis_is_primary_disease"
## [9] "ajcc_pathologic_t"        "morphology"
## [11] "classification_of_tumor"   "sites_of_involvement"
## [13] "days_to_last_follow_up"   "follow_ups_disease_response"
## [15] "race"                     "gender"
## [17] "ethnicity"                 "vital_status"
## [19] "age_at_index"              "days_to_birth"
## [21] "age_is_obfuscated"         "bcr_patient_barcode"
## [23] "primary_site"              "disease_type"
```

`clinical_data` is a table with 1231 rows and 24 columns:

- The rows correspond to the names of the patients, similar to `GeneX`.
- The columns are the names of the clinical data, such as `initial_weight`, `sites_of_involvement`, and `vital_status`.
- Each cell corresponds to the value of the clinical data for each patient, which can be either an integer or a character string.

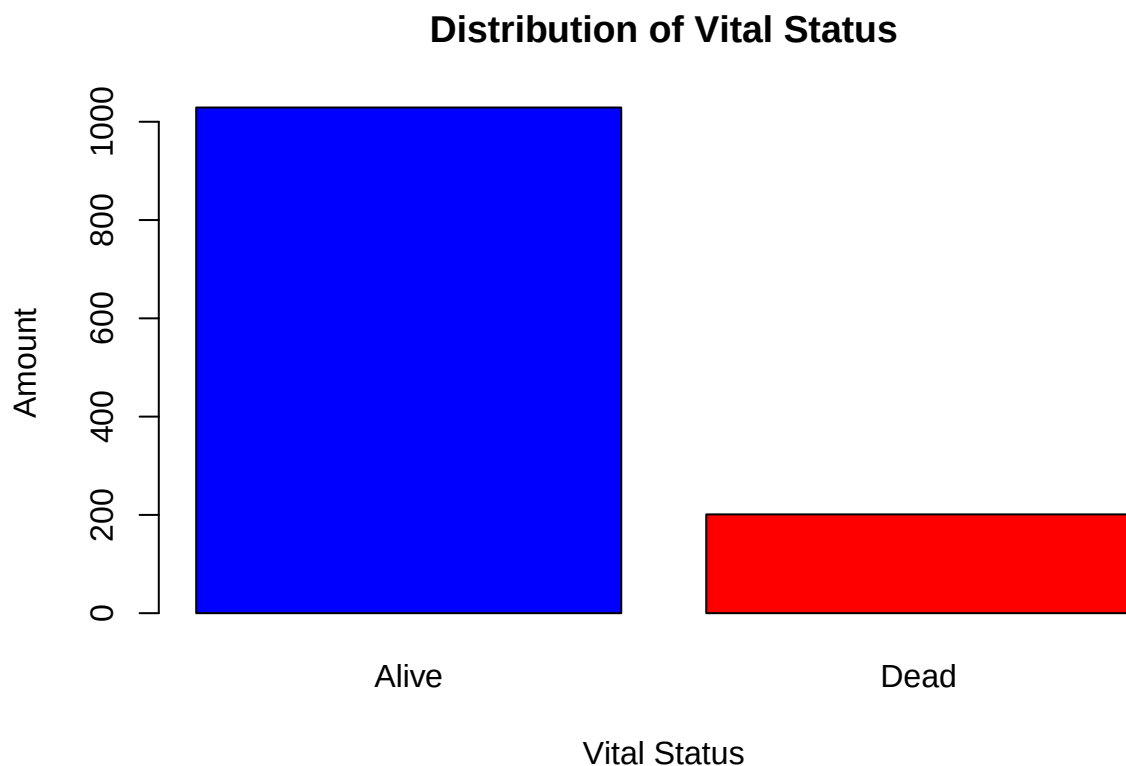
Outcome variable

The outcome variable is the vital status of the patient (dead or alive).

```
y <- clinical_data$vital_status  
table(y)
```

```
## y  
## Alive Dead  
## 1029 201
```

```
barplot(table(y),  
        main = "Distribution of Vital Status",  
        xlab = "Vital Status",  
        ylab = "Amount",  
        col = c("blue", "red"))
```



Exploratory data analysis

Genes

For the genes table, let's first plot for each gene the gaussian distribution of proportion of dead minus proportion of alive and then take the average of this graph.

```
# Calculate mean expression for each gene by vital status  
for (col in colnames(GeneX)[1:5]) { # First 5 genes for demonstration  
  gene_data <- GeneX[, col]  
  status <- clinical_data$vital_status
```

```

# Split data by status
alive_expr <- gene_data[status == "Alive"]
dead_expr <- gene_data[status == "Dead"]

# Calculate means
mean_alive <- mean(alive_expr, na.rm = TRUE)
mean_dead <- mean(dead_expr, na.rm = TRUE)

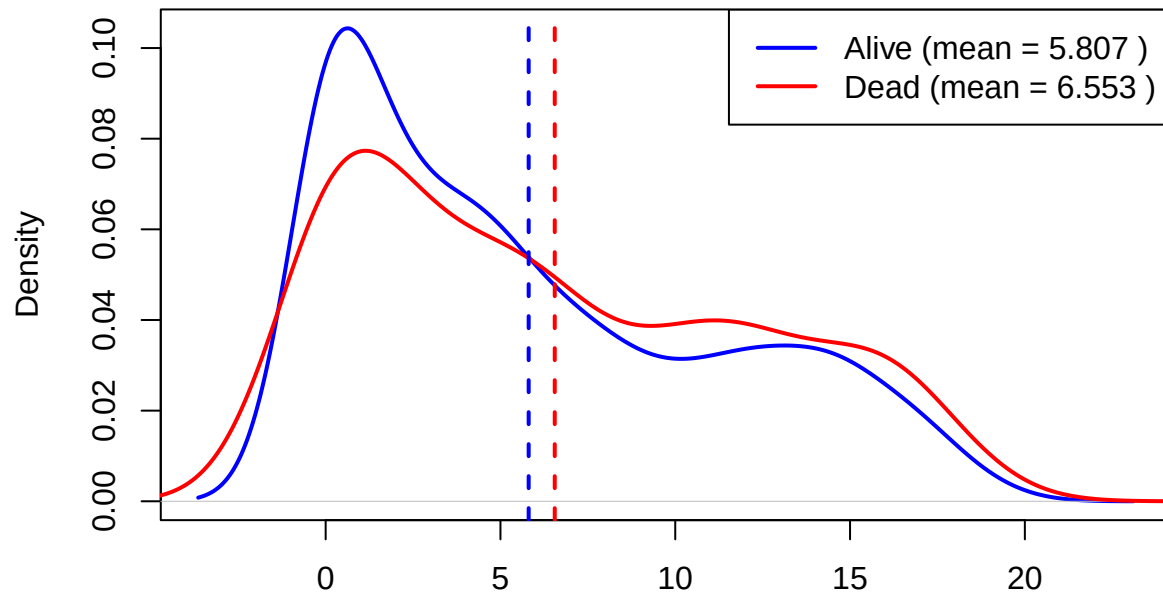
# Create density plot
plot(density(alive_expr, na.rm = TRUE),
     col = "blue",
     lwd = 2,
     main = paste("Gene:", col),
     xlab = "Expression Level",
     ylab = "Density")
lines(density(dead_expr, na.rm = TRUE),
     col = "red",
     lwd = 2)

# Add mean lines
abline(v = mean_alive, col = "blue", lty = 2, lwd = 2)
abline(v = mean_dead, col = "red", lty = 2, lwd = 2)

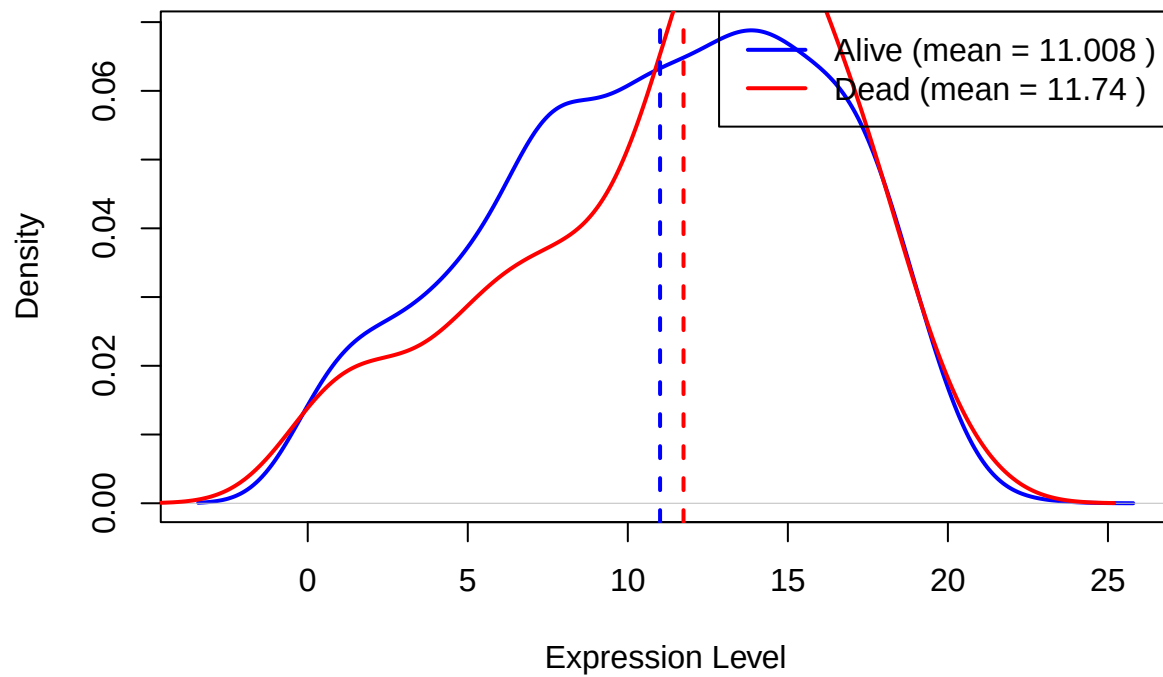
# Add legend
legend("topright",
      legend = c(paste("Alive (mean =", round(mean_alive, 3), ")"),
                  paste("Dead (mean =", round(mean_dead, 3), ")")),
      col = c("blue", "red"),
      lty = 1,
      lwd = 2)
}

```

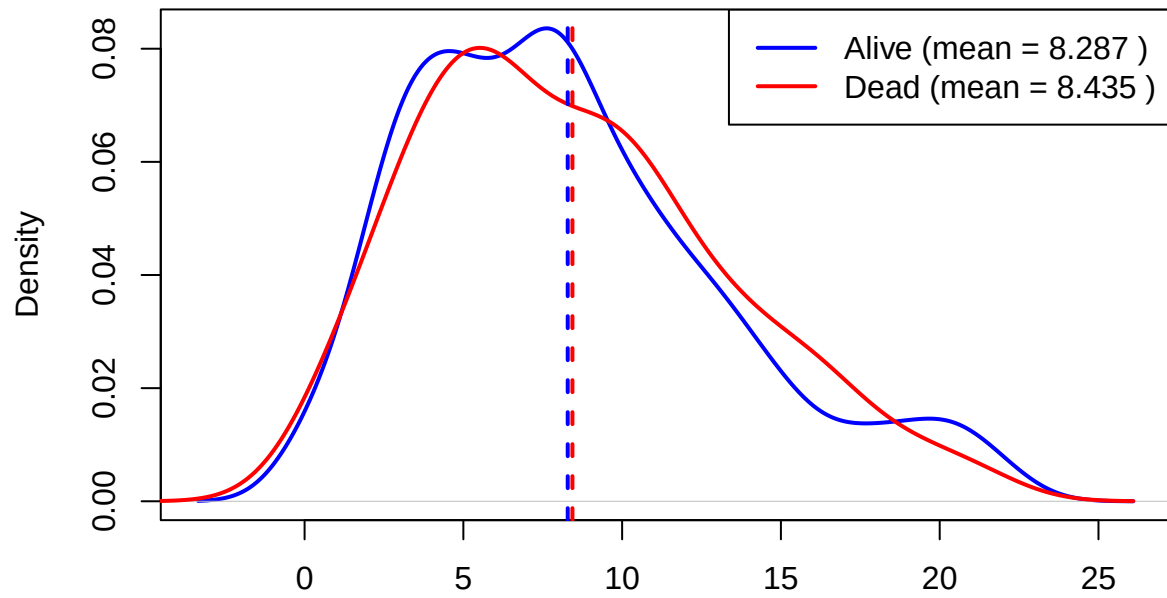
Gene: CLEC3A



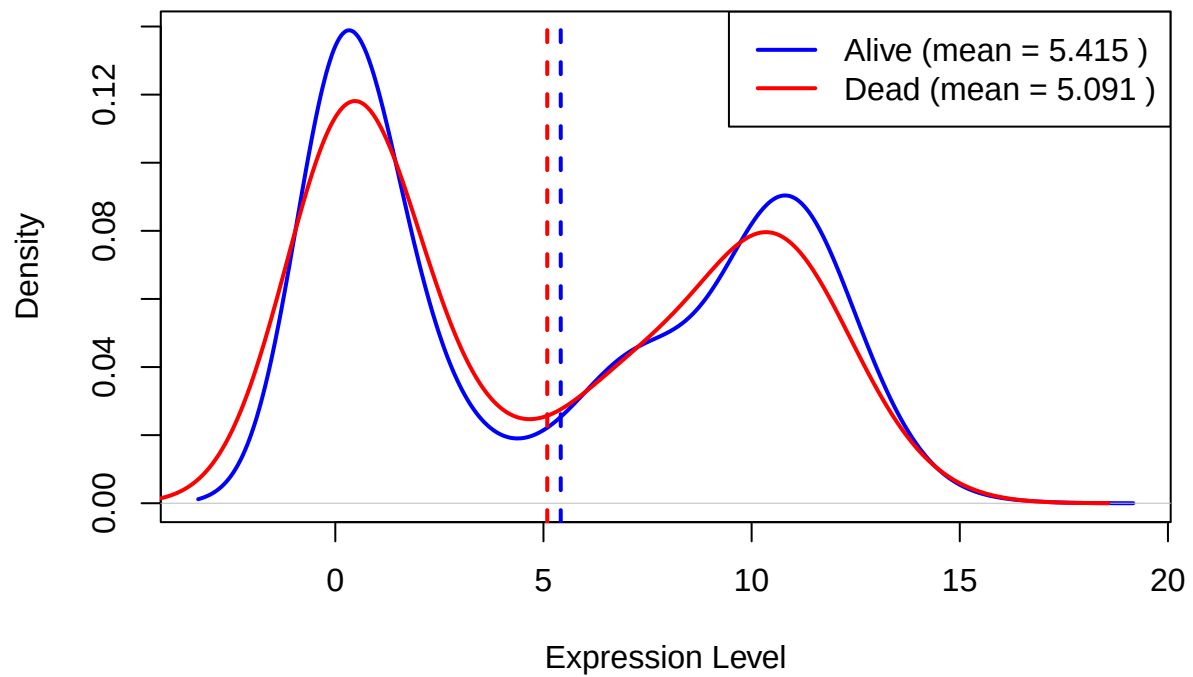
Gene: SCGB2A2



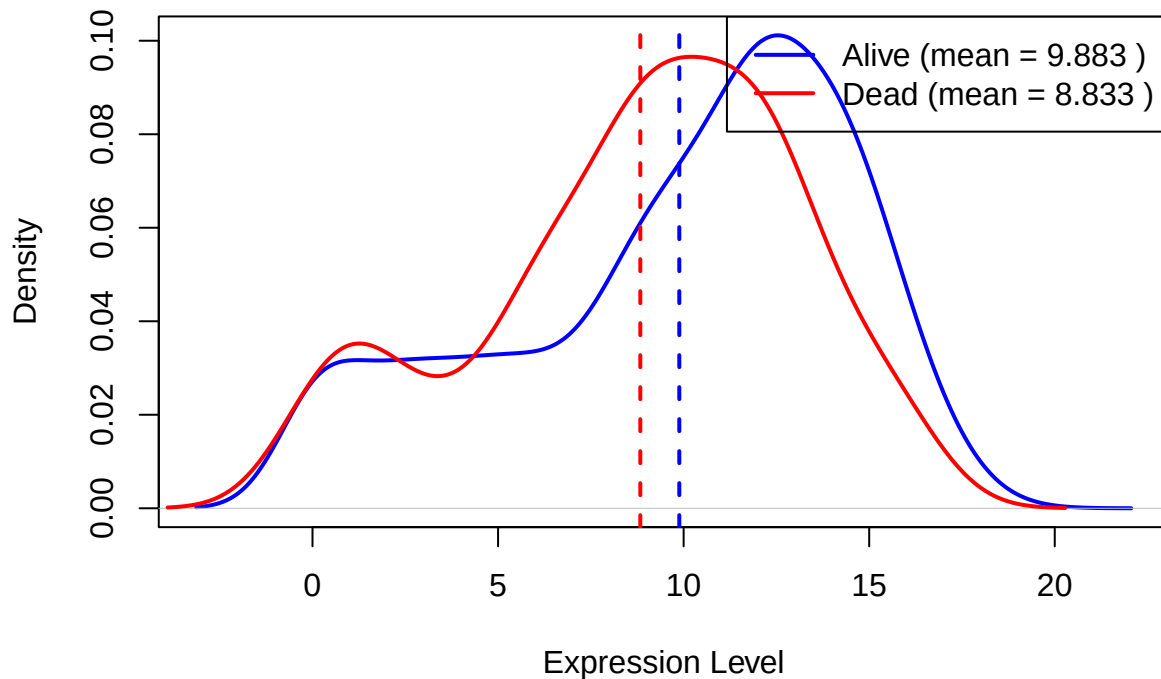
Gene: CPB1



Gene: GSTM1



Gene: TFF1



Thus we can display the most important genes by calculating the relative mean difference between the two groups (alive and dead) for each gene and then sorting them.

```
gene_diff <- data.frame(
  Gene = colnames(GeneX),
  MeanDiff = numeric(ncol(GeneX))
)

for (i in seq_along(colnames(GeneX))) {
  col <- colnames(GeneX)[i]
  gene_data <- GeneX[, col]
  status <- factor(clinical_data$vital_status)

  plot_data <- data.frame(Expression = gene_data, Status = status)
  means <- aggregate(Expression ~ Status, data = plot_data, mean)

  # Calculate relative mean difference
  mean_alive <- means$Expression[means$Status == "Alive"]
  mean_dead <- means$Expression[means$Status == "Dead"]

  # Relative difference: (Dead - Alive) / Alive
  # Handle division by zero
  if (mean_alive != 0) {
    gene_diff$MeanDiff[i] <- abs((mean_dead - mean_alive) / mean_alive)
  } else {
    gene_diff$MeanDiff[i] <- NA
  }
}

# Sort by relative mean difference
```

```

gene_diff <- gene_diff[order(gene_diff$MeanDiff, decreasing = TRUE), ]

# Display top 50 genes
top_50 <- head(gene_diff, 50)
print(top_50)

```

```

##           Gene MeanDiff
## 900      LALBA 0.8992166
## 1506     SULT1C3 0.7832339
## 2589      CSN2 0.7646311
## 4892    LINC01497 0.7279608
## 3569    AC007423.1 0.6802403
## 2308      RPS4Y1 0.5873672
## 2128     HSD3BP2 0.5872528
## 737     AC104407.1 0.5689319
## 1657    FP325317.1 0.5546457
## 2995      LHCGR 0.5456845
## 3509    LINC02237 0.5450194
## 3699      NGF-AS1 0.5247317
## 4982    LINC01612 0.5070537
## 748      CSN1S1 0.4944819
## 2205    ALDH1L1-AS2 0.4744773
## 3488      EPB42 0.4720326
## 1133    AC008459.1 0.4695645
## 192      NPY2R 0.4656152
## 3270    AL121950.1 0.4547892
## 3963    AL353693.1 0.4437907
## 4992    NEUROG2-AS1 0.4373508
## 4097      SGCZ 0.4328951
## 1595      SYT4 0.4293564
## 3806      SERTM1 0.4274395
## 3696    AC093496.1 0.4251907
## 587      SMR3B 0.4246711
## 3893    AP001360.1 0.4203018
## 3438    LINC01456 0.4180961
## 4182    AC084212.1 0.4096885
## 1784    AL356218.2 0.4084072
## 2672    AL591686.1 0.4064550
## 2057    LINC02511 0.4056917
## 743      APOB 0.4002554
## 2201      ADH4 0.3999677
## 3164    LINC01186 0.3998710
## 3053      PGM5-AS1 0.3971454
## 2269      PROKR1 0.3952935
## 974      NEUROG2 0.3935187
## 1003    LINC01667 0.3928182
## 2475      ADH1A 0.3913356
## 525      MYOC 0.3877588
## 2289    AL845331.1 0.3750996
## 2805    AC073850.1 0.3730908
## 1092    LINC00668 0.3683282
## 3341      BPIFA1 0.3681901
## 4672    LINC01322 0.3680301
## 1430    AC002546.1 0.3669965

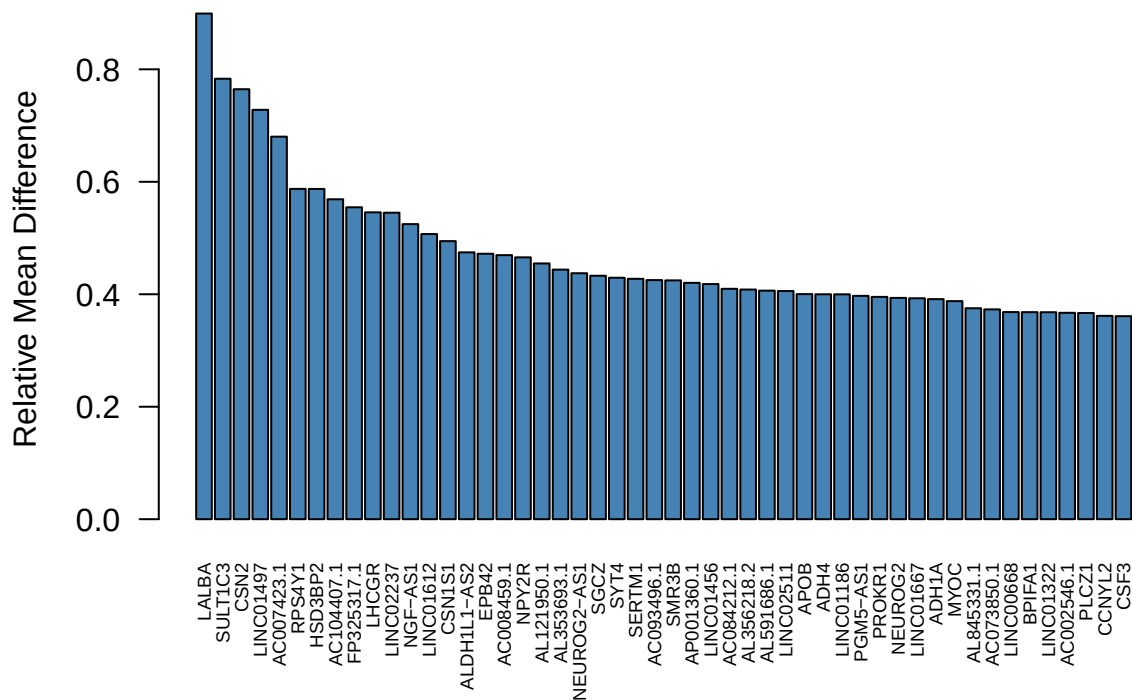
```

```
## 3273      PLCZ1 0.3666600
## 4006      CCNYL2 0.3616090
## 1772      CSF3 0.3610313
```

```
# Optional: visualize the distribution
```

```
barplot(top_50$MeanDiff[1:50],
        names.arg = top_50$Gene[1:50],
        las = 2,
        cex.names = 0.6,
        main = "Top 50 Genes by Relative Mean Difference",
        ylab = "Relative Mean Difference",
        col = "steelblue")
```

Top 50 Genes by Relative Mean Difference



Clinical Data

First modeling attempt with three variables from clinical data: `initial_weight`, `classification_of_tumor`, `morphology` to predict `vital_status`.

```
x <- clinical_data[, c("initial_weight",
                       "classification_of_tumor",
                       "morphology")]
y <- clinical_data$vital_status
model_data <- data.frame(x, vital_status = factor(y))
model_data <- na.omit(model_data) # removes NA for now
model_data$vital_status <- droplevels(model_data$vital_status)
```

Model fitting:

```
model <- glm(vital_status ~ ., data = model_data, family = binomial)
```

Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

Prediction:

```
pred <- predict(model, type = "response")
positive_class <- levels(model_data$vital_status)[2]
pred_class <- ifelse(pred > 0.5, positive_class, levels(model_data$vital_status)[1])
pred_class <- factor(pred_class, levels = levels(model_data$vital_status))
```

Confusion matrix:

```
conf_mat <- table(Predicted = pred_class, Observed = model_data$vital_status)
conf_mat
```

```
##           Observed
## Predicted Alive Dead
##      Alive   997  150
##      Dead    20   48
```

Visualization of the confusion matrix.

```
conf_mat_matrix <- as.matrix(conf_mat)

image(t(conf_mat_matrix[nrow(conf_mat_matrix):1, ]),
      axes = FALSE,
      col = colorRampPalette(c("#f7fbff", "#6baed6", "#08306b"))(64),
      xlab = "Observed",
      ylab = "Predicted",
      main = "Confusion Matrix")

axis(1, at = seq(0, 1, length.out = ncol(conf_mat_matrix)), labels = colnames(conf_mat_matrix))

axis(2, at = seq(0, 1, length.out = nrow(conf_mat_matrix)), labels = rev(rownames(conf_mat_matrix)))

text(rep(seq(0, 1, length.out = ncol(conf_mat_matrix)), each = nrow(conf_mat_matrix)),
     rep(seq(0, 1, length.out = nrow(conf_mat_matrix)), times = ncol(conf_mat_matrix)),
     labels = as.vector(conf_mat_matrix[rev(seq_len(nrow(conf_mat_matrix))), ]),
     col = "black", font = 2)
```

Confusion Matrix

